Benchmarking Classical and Quantum Classifiers for Resting-State EEG: Pretty Good Measurement, QAOA-CV and Kernel Learning

Gregoire Cattan, Alexandre Quemy, and Ulvi Movsum-zada

***Abstract*—Quantum-inspired and quantum machine learning (QML) methods have attracted considerable interest in neurotechnology, yet their benefit over classical baselines on realistic brain–computer interface (BCI) data remains unclear. This paper benchmarks thirteen classical and quantum pipelines on three resting-state EEG datasets from the MOABB framework under a cross-subject evaluation protocol with grouped k-fold splitting. We introduce three contributions evaluated for the first time on resting-state EEG data: (i) QUANTUMSTATEDISCRIMINATOR, a novel quantum-inspired classifier that models mental states as density matrices, constructs a Positive Operator-Valued Measure (POVM) via the Pretty Good Measurement (PGM), and scores test trials using the Born rule—entirely bypassing the explicit covariance estimation step; (ii) the *Ulvi* optimizer variant of Quantum Approximate Optimization with Continuous Variables (QAOA-CV) with circular entanglement for the Nearest Convex Hull (NCH) classifier; and (iii) Intrinsic Green's Learning (IGL), a PDE-inspired kernel framework applied to raw EEG time series. Experiments show that QUANTUMSTATEDISCRIMINATOR and QIOCE achieve mean ROC AUC scores comparable to the best classical baseline (MDM: 0.613), reaching 0.628 and 0.626, respectively, while the QUANTUMSTATEDISCRIMINATOR is the fastest classifier tested at under 0.3 s mean training time. NCH-based quantum methods trail classical baselines but benefit from Riemannian transfer learning in synthetic domain-shift conditions. IGL underperforms on these datasets. These findings provide a rigorous first benchmark for the three proposed methods and highlight the trade-off between quantum-algorithmic expressiveness and practical performance on near-term simulated quantum hardware.**

***Index Terms*—brain–computer interface, resting-state EEG, quantum machine learning, Pretty Good Measurement, QAOA, transfer learning, MOABB, Riemannian geometry**

I. INTRODUCTION

Electroencephalography (EEG)-based brain–computer interfaces (BCIs) enable direct communication between users and devices by decoding neural activity. Resting-state paradigms—where participants alternate between distinct mental states such as relaxed rest and passive task—are of particular clinical interest because they impose minimal cognitive demands and are applicable to populations unable to perform active motor imagery [1]–[3]. However, the non-stationarity of EEG and large between-subject variability make cross-subject generalization a persistent challenge [4].

Riemannian geometry has emerged as the dominant framework for EEG classification [5], treating covariance matrices as points on the symmetric positive definite (SPD) manifold and applying geodesic distances for nearest-mean classification (MDM) or projecting to the tangent space for linear discriminants [6]. Transfer learning (TL) extensions—Riemannian Procrustes Analysis (RPA)—align subject covariance distributions through centering, scaling, and rotation [7], improving cross-subject performance.

Quantum and quantum-inspired machine learning has been proposed for biosignal classification, motivated by the structural similarity between density matrices and EEG covariance matrices [8]. The Quantum Approximate Optimization Algorithm (QAOA) [9] has been adapted to continuous-variable (CV) settings [10] and applied to EEG data through the Nearest Convex Hull classifier [11]. The Pretty Good Measurement [12] offers a principled quantum-theoretic framework for optimal state discrimination.

Despite this activity, no study has systematically compared quantum and classical EEG classifiers under a rigorous cross-subject benchmark with standardized evaluation. This paper addresses that gap with the following contributions:

1) **QUANTUMSTATEDISCRIMINATOR**: a quantum-inspired classifier that maps raw EEG epochs directly to density matrices and performs optimal quantum state discrimination via the PGM without requiring an explicit Riemannian covariance step.
2) **NCH+QAOA-CV (Ulvi optimizer)**: first application of the Ulvi circular-entanglement QAOA-CV optimizer to EEG classification, and evaluation of the effect of transfer learning alignment on NCH.
3) **IGL on resting-state EEG**: first benchmark of Intrinsic Green's Learning—a PDE-inspired adaptive kernel method—on EEG mental state classification.
4) **Cross-subject TL infrastructure for MOABB**: the `Adapter` meta-estimator and `TLCrossSubjectEvaluation` class enable seamless integration of pyriemann transfer-learning pipelines into MOABB's cross-subject evaluation loop, forwarding subject IDs and target domain at fit time without modifying the pipeline API.

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All pipelines are evaluated on three MOABB resting-state datasets under identical cross-subject conditions, enabling fair head-to-head comparison. The remainder of this paper is organized as follows. Section II reviews the necessary background. Section III describes all pipelines and the evaluation protocol. Section IV reports benchmark results. Section V presents controlled ablation studies. Section VI discusses findings, and Section VII concludes.

II. Background

A. Riemannian EEG Processing

Given a raw EEG epoch $X \in \mathbb{R}^{C\times T}$ (C channels, T time samples), the sample covariance matrix is

$$\Sigma = \frac{1}{T-1} X X^\top \in \mathcal{S}_C^+, \tag{1}$$

where $\mathcal{S}_C^+$ denotes the manifold of $C\times C$ symmetric positive definite (SPD) matrices.

In practice, the sample covariance is ill-conditioned when C is large relative to T. The Ledoit–Wolf (LWF) shrinkage estimator [13] regularizes toward a scaled identity:

$$\hat{\Sigma}_{\text{lwf}} = (1-\alpha)\,\Sigma + \alpha\,\frac{\operatorname{tr}(\Sigma)}{C}\,I, \tag{2}$$

where $\alpha \in [0,1]$ is chosen analytically to minimize the expected squared Frobenius error between the estimator and the true covariance. All pipelines requiring covariance matrices use the LWF estimator throughout this paper.

a) MDM.: The Minimum Distance to Mean (MDM) classifier [5] assigns a test covariance Σ to the class whose Riemannian mean μ_c is closest:

$$\hat{y} = \arg\min_c \; \delta_R(\Sigma, \mu_c), \tag{3}$$

where $\delta_R(A,B) = \|\log(A^{-1/2}BA^{-1/2})\|_F$ is the affine-invariant Riemannian distance.

b) Tangent Space + LDA.: Covariance matrices are projected to the tangent space at the Riemannian mean of training data, yielding Euclidean feature vectors on which a Linear Discriminant Analysis (LDA) classifier is trained [6].

B. Riemannian Transfer Learning

Riemannian Procrustes Analysis (RPA) [7] aligns covariance distributions across subjects through a sequence of operations:

- **TLCenter**: recenters each subject's distribution to the identity matrix.
- **TLScale**: normalizes trace variance across subjects.
- **TLRotate**: rotates subject manifolds to align class means.

A `TLClassifier` wrapper then trains a downstream estimator on the aligned, labeled data from multiple source subjects. Figure 7 illustrates these three operations geometrically on synthetic cross-subject EEG data.

C. QAOA and Continuous-Variable Extensions

The Quantum Approximate Optimization Algorithm (QAOA) [9] is a variational quantum algorithm of p alternating cost and mixer Hamiltonian layers, where the *cost operator is derived directly from the combinatorial objective function*. In the continuous-variable (CV) extension [10], displacement operators replace qubit gates and the cost Hamiltonian encodes a real-valued objective.

Three generations of QAOA-CV classifiers have been applied to EEG data:

1) **NaiveQAOA**: standard QAOA-CV where the cost operator directly encodes the NCH squared-distance objective. The *Nearest Convex Hull (NCH)* classifier [11] uses this ansatz to find, for each class, the point in the convex hull of tangent-space covariances closest to a test point, classifying by nearest hull.
2) **QAOA-CV** [14] (Luna et al.): keeps the same circuit structure and cost operator as NaiveQAOA (Ising Hamiltonian derived from the objective), but replaces the standard loss. Instead of minimizing the Hamiltonian expectation value $\langle\psi|H_P|\psi\rangle$, it evaluates the classical cost function directly at continuous-variable estimates recovered from the circuit's marginal measurement probabilities: each variable x_i is read out as $\hat{x}_i = \text{scaler}^{-1}(\Pr[\text{qubit}\, i = |0\rangle])$, and the loss is $\mathcal{L} = f(\hat{x}_1,\ldots,\hat{x}_n)$. This approach is therefore a variational algorithm with QAOA circuit structure but a non-Hamiltonian, problem-native loss.
3) **QIOCE** [15]: departs from classical QAOA by using a cost operator *not* derived from the objective. The circuit ansatz is:
$$\texttt{QAOAAnsatz}\begin{pmatrix} \text{cost} = R_x R_y R_z \;\forall \text{ variable}, \\ \text{reps} = 5, \\ \text{initial_state} = R_y \;\forall \text{ variable}, \\ \text{mixer} = \text{circular entanglement} \end{pmatrix}$$
with a custom loss function optimized by NFT. QIOCE is therefore better described as a parametric quantum circuit (PQC) classifier with QAOA-like layered structure than as a strict QAOA instance.

D. Pretty Good Measurement

Given K quantum states $\{\rho_k\}$ with prior probabilities $\{\pi_k\}$, the Pretty Good Measurement (PGM) [12] constructs POVM elements

$$\Pi_k = \rho_{\text{tot}}^{-1/2}\,(\pi_k\rho_k)\,\rho_{\text{tot}}^{-1/2}, \qquad \rho_{\text{tot}} = \sum_k \pi_k\rho_k. \tag{4}$$

By construction, $\sum_k \Pi_k = I$, so Born-rule scores $p(k \mid \rho) = \operatorname{tr}(\Pi_k\rho)$ are valid probabilities. For two equiprobable pure states, the PGM achieves the Helstrom bound [16].

E. MOABB and Cross-Subject Evaluation

The Mother of All BCI Benchmarks (MOABB) [17] provides a standardized framework for reproducible EEG decoding experiments. We use the `CrossSubjectEvaluation` protocol extended with grouped k-fold splitting, where each fold uses a disjoint set of source subjects for training and a held-out subject for testing.

III. Methods

A. Datasets

Three MOABB resting-state datasets are used (Table I):

TABLE I
DATASETS USED IN THE BENCHMARK.

Dataset	Subjects	Ch	Task	Classes
Cattan2019_PHMD [1]	12	16	PHMD rest	on/off
Rodrigues2017 [2]	50	16	Eyes open/closed	closed/open
Hinss2021 [3]	15	32	Mental workload	easy/medium

Epochs are extracted using the `RestingStateToP300Adapter` paradigm. For Hinss2021, epochs span 0–0.5 s ($T_{\max} = 0.5$ s). The Ledoit–Wolf regularized covariance estimator (`lwf`) is used for all pipelines that require covariance matrices.

B. Evaluation Protocol

All evaluations use `TLCrossSubjectEvaluation`, a subclass of `CrossSubjectEvaluation` that: (i) passes group labels (subject IDs) to transfer-learning pipelines; and (ii) sets the alignment target domain to the first training subject in each fold. Evaluation uses GroupKFold with $k = 3$ splits and ROC AUC as the performance metric.

C. Classical Baselines

a) MDM.: Ledoit–Wolf covariance → MDM [5].

b) TS+LDA.: Ledoit–Wolf covariance → TangentSpace (δ_R) → LDA [6].

c) MDWM(0.5).: Domain-weighted minimum distance to mean with equal weighting between source and target domains (domain tradeoff $= 0.5$) [18].

d) MDM+TL / TS+LDA+TL.: Full RPA alignment (TLCenter → TLScale → TLRotate) prepended to MDM or TS+LDA [7].

D. Quantum Baselines

a) NCH+NaiveQAOA (±TL).: Ledoit–Wolf covariance → QuanticNCH with standard QAOA-CV (NaiveQAOA), $p = 2$ layers, subsampling to $n = 6$ hull points per class [11].

b) QIOCE (±TL).: Ledoit–Wolf covariance → Whitening (4 components) → TangentSpace → ContinuousQIOCEClassifier, $p = 2$ layers, NFT optimizer (25 iterations), max 10 features [15].

E. Novel Contributions

1) QuantumStateDiscriminator: The brain state (rest, task) is modelled as a *mixed quantum state* (density matrix ρ_c)—a generalisation of a probability distribution over possible neural configurations. Each EEG recording is treated as a *measurement operator* M, and classification reduces to identifying which class quantum state ρ_c best matches the observed measurement via the Born rule. Crucially, this bypasses the Riemannian covariance pipeline entirely: raw epochs feed directly into the classifier.

The QuantumStateDiscriminator learns one density matrix per class via quantum state tomography on labelled training EEG, then constructs a Positive Operator-Valued Measure (POVM) using the Pretty Good Measurement. Table II summarises the mapping between quantum concepts and EEG quantities.

TABLE II
QUANTUM MECHANICS TO EEG IMPLEMENTATION MAPPING.

Quantum concept	EEG implementation
System state (mixed)	Class density matrix ρ_c (learned)
Observable / measurement	$M = XX^\top/T$, normalised to $\mathrm{tr} = 1$
Class prior	$\pi_c = N_c/N_{\text{tot}}$
POVM element	$\Pi_c = \rho_{\text{tot}}^{-1/2}(\pi_c\rho_c)\rho_{\text{tot}}^{-1/2}$
Born rule score	$\mathrm{tr}(\Pi_c \cdot M) \in [0, 1]$, sums to 1

a) Fit: quantum state tomography.: For each class $c \in \{1, \ldots, K\}$ with N_c training trials:

$$\Sigma_c = \frac{1}{N_c} \sum_{i:y_i=c} \frac{X_i X_i^\top}{T_i}, \tag{5}$$

$$\rho_c = \frac{\Sigma_c}{\mathrm{tr}(\Sigma_c)}, \tag{6}$$

$$\pi_c = \frac{N_c}{N_{\text{total}}}. \tag{7}$$

The EEG measurement operator Σ_c is the mean sample covariance (unnormalized SCM); ρ_c is its trace-normalization to a valid density matrix.

b) Fit: Pretty Good Measurement.: Compute the prior-weighted average state and its inverse square root:

$$\rho_{\text{tot}} = \sum_c \pi_c \rho_c, \tag{8}$$

$$\rho_{\text{tot}}^{-1/2} = V \ \mathrm{diag}(1/\sqrt{\lambda_i})\, V^\top \quad \text{(eigendecomposition)}, \tag{9}$$

then construct the POVM:

$$\Pi_c = \rho_{\text{tot}}^{-1/2} \left(\pi_c \rho_c\right) \rho_{\text{tot}}^{-1/2}. \tag{10}$$

By construction $\sum_c \Pi_c = I$, so POVM scores are valid probability distributions.

c) Predict: Born rule.: For a new EEG trial X, build the normalized measurement operator

$$M = \frac{XX^\top/T}{\mathrm{tr}(XX^\top/T)}, \tag{11}$$

and compute the Born-rule score for each class:

$$\text{score}_c = \mathrm{tr}(\Pi_c \cdot M) = \sum_{ij} (\Pi_c)_{ij} M_{ij}. \tag{12}$$

Predicted class: $\hat{y} = \arg\max_c \ \text{score}_c$. Scores are returned directly as probabilities; no softmax is applied.

d) Comparison with MDM.: Table III contrasts QuantumStateDiscriminator with Covariances+MDM. Both are nearest-mean classifiers but differ in feature normalization, similarity measure, and how classes are coupled.

The inter-class coupling through ρ_{tot} makes QuantumStateDiscriminator closer in spirit to LDA than to MDM, despite the name. A true quantum MDM would compute

TABLE III
COMPARISON OF MDM AND QUANTUMSTATEDISCRIMINATOR.

	Cov + MDM	QuantumStateDiscriminator
Feature	$XX^\top/(T-1)$	$XX^\top/T$, norm. to $\mathrm{tr}=1$
Class prototype	Riemannian mean μ_c	Density matrix ρ_c
Similarity	Riemannian distance	Quantum fidelity (PGM)
Class coupling	Independent	Coupled via ρ_{tot}
Priors	Not used	$\pi_c = N_c/N_{\text{tot}}$
Covariance step	Required	Not required

Algorithm 1 QUANTUMSTATEDISCRIMINATOR

Input: training set $\{(X_i, y_i)\}$, test epoch X

for each class c **do**
  $\Sigma_c \leftarrow \frac{1}{N_c}\sum_{i:y_i=c} X_i X_i^\top / T_i$
  $\rho_c \leftarrow \Sigma_c / \mathrm{tr}(\Sigma_c)$; $\pi_c \leftarrow N_c/N_{\text{tot}}$
end for
$\rho_{\text{tot}} \leftarrow \sum_c \pi_c \rho_c$
$\rho_{\text{tot}}^{-1/2} \leftarrow$ eigendecomposition of ρ_{tot}
for each class c **do**
  $\Pi_c \leftarrow \rho_{\text{tot}}^{-1/2}(\pi_c\rho_c)\rho_{\text{tot}}^{-1/2}$
end for
$M \leftarrow XX^\top/T / \mathrm{tr}(XX^\top/T)$
return $\arg\max_c \ \mathrm{tr}(\Pi_c \cdot M)$

$\arg\min_c \delta_Q(\rho_M, \rho_c)$ using e.g. the Bures distance or quantum relative entropy without a POVM; the present classifier is more precisely a *quantum state discrimination* method.

e) Regularization note.: The measurement operator $M = XX^\top/T/\mathrm{tr}(XX^\top/T)$ is the trace-normalised biased SCM. No regularization is applied, which can lead to ill-conditioning when the number of channels C is large relative to the number of time samples T. A natural extension is to apply shrinkage (e.g. Ledoit–Wolf) before trace normalization.

2) NCH+QAOA-CV (Ulvi Optimizer): The Ulvi variant of QAOA-CV introduces circular entanglement between qumodes via beam-splitter gates arranged in a ring topology, alternating with single-mode displacement and phase-shift operators. The mixer is constructed using `create_mixer_with_circular_entanglement(0)`. All other hyperparameters match the NaiveQAOA baseline: $p = 2$ layers, NFT optimizer (25 iterations), $n = 6$ hull points per class, `shots` $= 100$.

3) IGL: Intrinsic Green's Learning: IGL [19] (Quemy, ICLR 2026) models the target function as the solution to a linear PDE $\mathcal{L}u = f$, where $\mathcal{L}$ is a differential operator chosen as an architectural inductive bias (e.g. Laplacian for smooth fields, Helmholtz for localized features) and f is a *learned source term*. The solution is expressed via Green's function:

$$u(\xi) = (\mathcal{L}^{-1}f)(\xi) = \int_{\mathcal{M}} G(\xi,\zeta)\, f(\zeta)\, \mathrm{d}\zeta. \tag{13}$$

Rather than approximating u directly, IGL learns f and integrates it against G—integration is a stable operation (unlike differentiation) and makes u linear in the source weights, enabling a near-convex Stage 2 solve.

a) Factorized integral.: Both the source and the kernel are parameterized as low-rank tensors. The source is a rank-R CP decomposition and the Green's kernel is a rank-K product of univariate kernels:

$$\hat{f}(\zeta) = \sum_{r=1}^{R} w_r \prod_{j=1}^{d} \phi_{r,j}(\zeta^j), \tag{14}$$

$$G(\xi,\zeta) \approx \sum_{k=1}^{K} \gamma_k \prod_{j=1}^{d} G_{k,j}(\xi^j,\zeta^j). \tag{15}$$

By Fubini's theorem, the d-dimensional integral in (13) factorizes into d independent 1D integrals:

$$u(\xi) = \sum_{k,r} \gamma_k w_r \prod_{j=1}^{d} I_{k,r,j}(\xi^j), \quad I_{k,r,j}(\xi^j) = \int G_{k,j}(\xi^j,\zeta^j)\, \phi_{r,j}(\zeta^j)\, \mathrm{d}\zeta^j \tag{16}$$

at total cost $\mathcal{O}(DW^2 + KRd)$ per sample, where D is the ambient dimension and $d \ll D$ is the intrinsic dimension. This is D-independent at Stage 2, in contrast to $\mathcal{O}(N^2)$ for kernel methods or $\mathcal{O}(DW^2)$ for MLPs. An encoder $\Psi : \mathbb{R}^D \to \mathbb{R}^d$ discovers the coordinate chart $\xi = \Psi(x)$ where the source admits low-rank structure; learnable Group-Lasso gates on each coordinate dimension automatically discover the effective intrinsic dimension $d_{\text{eff}} \le d_{\max}$ [20].

b) Time-series variant (our adaptation).: The original IGL operates on flat feature vectors $X \in \mathbb{R}^D$: an MLP encoder discovers a low-dimensional latent chart $\xi = \Psi(x) \in \mathbb{R}^d$ ($d \ll D$), Hard Concrete Gates select the effective intrinsic dimension d_{eff}, and the Green's kernel integrates over ξ-space.

For raw EEG epochs $X \in \mathbb{R}^{C\times T}$ we use `IGLTimeSeriesSklearnClassifier`, which departs from this in two fundamental ways:

1) **Time is the fixed intrinsic coordinate.** The PDE domain is the 1D time axis $t \in [0,1]^T$, so $d = 1$ by construction. No encoder or Hard Concrete Gates are needed: the intrinsic dimension is known *a priori* from the signal structure.
2) **Spatial filtering replaces the MLP encoder.** A linear map $W_s : \mathbb{R}^C \to \mathbb{R}^K$ acts as a spatial filter bank (analogous to CSP), producing K source signals $f_k(t) = W_s x(t)$ from the C-channel EEG at each time point.

The Green's kernel then integrates each source signal over time, evaluating the solution at the *final* time point: $u_k = \sum_s G(T,s)\, f_k(s)$, where $G(T,s) = \Phi[-1]^\top \Phi[s]$. With a Gaussian operator, $G(T,s)$ decays with temporal distance from T, so recent events are weighted more heavily than distant ones. The K-dimensional embedding $\mathbf{u}$ is mapped to class logits by a linear head W, and no covariance estimation is required.

c) Two-stage (VP) training.: Variable Projection [21] separates coordinate discovery from source fitting:

- **Stage 1 (outer, nonlinear)**: gradient descent on the encoder and kernel parameters.
- **Stage 2 (inner, linear)**: $W \leftarrow \arg\min_W \|\Phi W - y\|^2$ (Tikhonov least-squares, solved analytically).

By the envelope theorem, solving Stage 2 to optimality at each outer step forces Stage 1 gradients to reflect coordinate quality alone, preventing the dimensional collapse ($d_{\text{eff}} \to 0$) that occurs with joint gradient descent.

We use Gaussian operator, $K = 16$ spatial components, $R = 64$ anchor points, 3 scales, 500 epochs (100 warmup).

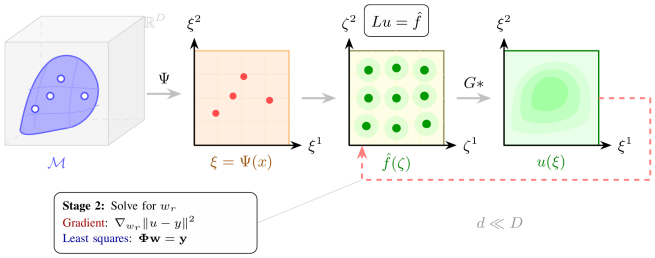

Fig. 1. IGL pipeline as a two-stage algorithm (reproduced from [19]). **Stage 1** (coordinate discovery): encoder Ψ maps data from the manifold $\mathcal{M} \subset \mathbb{R}^D$ to intrinsic coordinates $\xi \in \mathbb{R}^d$ where the tensor source $\hat{f}(\zeta) = \sum_r w_r \prod_j \phi_{r,j}(\zeta^j)$ decomposes. Green's convolution $G * \hat{f}$ yields the solution $u(\xi)$. **Stage 2** (source fitting): source weights w_r are solved via least squares ($\mathbf{\Phi w} = \mathbf{y}$).

F. Summary of Pipelines

Table IV lists all 13 evaluated pipelines.

TABLE IV
ALL EVALUATED PIPELINES.

Pipeline	TL	Type
MDM	–	Classical
MDM+TL	RPA	Classical
TS+LDA	–	Classical
TS+LDA+TL	RPA	Classical
MDWM(0.5)	DW	Classical
NCH+NaiveQAOA	–	Quantum
NCH+NaiveQAOA+TL	RPA	Quantum
NCH+QAOACV(Ulvi)	–	Quantum
NCH+QAOACV(Ulvi)+TL	RPA	Quantum
QIOCE	–	Quantum
QIOCE+TL	RPA	Quantum
QuantumStateDiscriminator	–	Quantum-inspired
IGL	–	Kernel

IV. RESULTS

A. Per-Pipeline AUC across All Datasets

Table V reports mean ROC AUC and mean training time (per fold, averaged across subjects and datasets) for all 13 pipelines.

Figure 2 shows the distribution of per-subject AUC scores across all three datasets. QUANTUMSTATEDISCRIMINATOR and QIOCE display higher mean performance and larger variability than MDM, while NCH variants cluster near chance level.

B. Statistical Analysis

Figure 3 shows the pairwise statistical comparison matrix produced by MOABB's `find_significant_differences` procedure. Cells report the standardized effect size and p-value from a Wilcoxon signed-rank test across datasets; green cells indicate significance at $\alpha = 0.05$. The effect size displayed is the standardized mean difference computed by MOABB's meta-analysis procedure (not Cohen's d in the strict parametric sense, since the Wilcoxon test is non-parametric).

TABLE V
MEAN ROC AUC AND TRAINING TIME ACROSS ALL THREE DATASETS.

Pipeline	AUC (mean)	Time (s, mean)
QuantumStateDiscriminator	**0.628**	**0.27**
QIOCE	**0.626**	933.3
MDM	0.613	16.5
MDWM(0.5)	0.591	36.7
MDM+TL	0.595	218.0
TS+LDA	0.584	27.3
NCH+QAOACV(Ulvi)	0.555	1.5
QIOCE+TL	0.572	821.8
NCH+NaiveQAOA	0.529	1.5
NCH+QAOACV(Ulvi)+TL	0.551	219.6
TS+LDA+TL	0.542	226.6
NCH+NaiveQAOA+TL	0.517	104.1
IGL	0.522	42.2

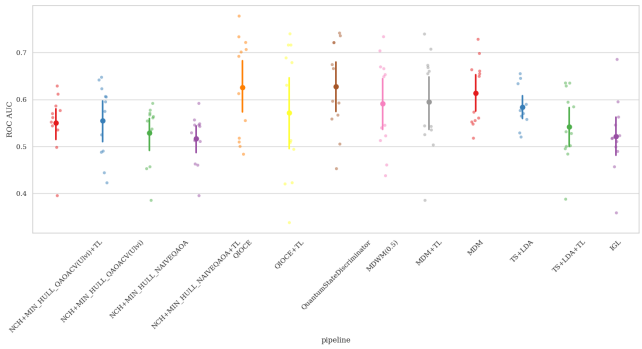

Fig. 2. ROC AUC distributions across all subjects and datasets. Strip plot (individual subjects, $\alpha = 0.5$) overlaid with mean $\pm$ 95% CI (point plot). Pipelines ordered by the preferred display order from the benchmark script.

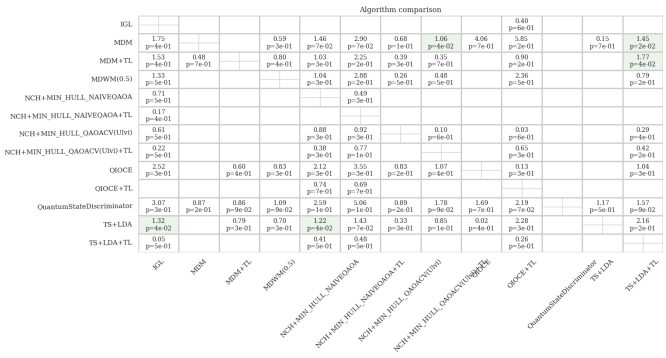

Fig. 3. Algorithm comparison matrix. Each cell shows the effect size and p-value for the row algorithm vs. the column algorithm. Green: row significantly better; red: column significantly better.

Key statistically significant differences ($p < 0.05$):

- **TS+LDA** > **IGL** ($p = 0.04$, SMD= 1.32): classical tangent-space projection outperforms IGL across all three

datasets (Fig. 4).

- **MDM+TL** $>$ **TS+LDA+TL** ($p = 0.04$, SMD$= 1.77$): after RPA alignment, MDM yields significantly higher AUC than TS+LDA.
- **MDM** $>$ **NCH+QAOACV(Ulvi)+TL** ($p = 0.04$): meta-analysis confirms consistent MDM advantage across datasets (Fig. 5).
- **TS+LDA** $>$ **NCH+NaiveQAOA** ($p = 0.04$): the classical tangent space baseline significantly outperforms Naive-QAOA (Fig. 6).

No statistically significant difference was detected between QUANTUMSTATEDISCRIMINATOR and MDM, nor between QIOCE and MDM.

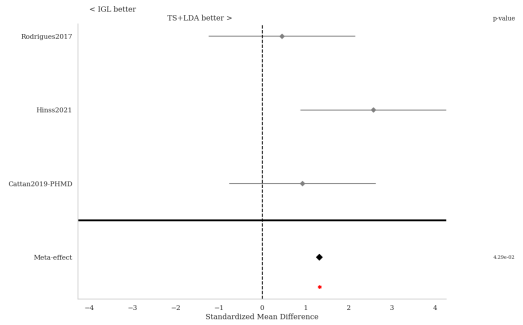

Fig. 4. Meta-analysis forest plot: TS+LDA vs. IGL. Positive SMD indicates TS+LDA is better. Meta-effect significant ($p = 0.043$).

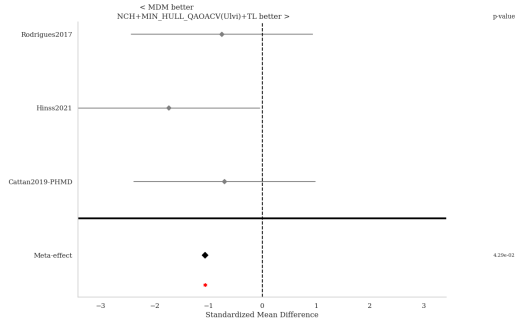

Fig. 5. Meta-analysis forest plot: MDM vs. NCH+QAOACV(Ulvi)+TL. Negative SMD indicates MDM is better. Meta-effect significant ($p = 0.043$).

C. Effect of Transfer Learning

Transfer learning consistently *hurts* QIOCE ($0.626 \rightarrow 0.572$), NCH+NaiveQAOA ($0.529 \rightarrow 0.517$), and TS+LDA ($0.584 \rightarrow 0.542$), while it has a small positive effect on MDM ($0.613 \rightarrow 0.595$; within noise) and leaves NCH+QAOACV(Ulvi) largely unchanged. The MDM+TL vs. TS+LDA+TL significance suggests that Riemannian MDM is more robust to the RPA alignment than the tangent-space projection.

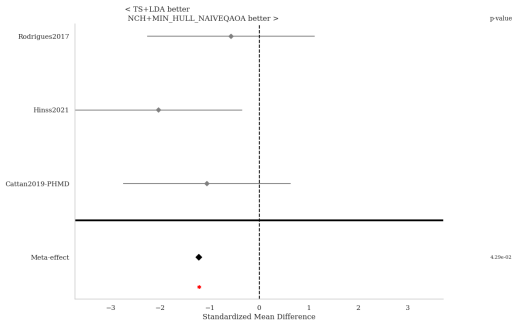

Fig. 6. Meta-analysis forest plot: TS+LDA vs. NCH+NaiveQAOA. Negative SMD indicates TS+LDA is better. Meta-effect significant ($p = 0.043$).

V. ABLATION STUDIES

A. Transfer Learning Benefit for NCH

To isolate the effect of RPA alignment on NCH, we generate synthetic EEG data with controlled domain shift: 10 simulated subjects, 4 channels, 50 time samples, 20 trials per class, with class-specific Cholesky mixing matrices to introduce per-subject domain offsets (see `noplot_nch_tl_ablation.py`).

Figure 7 illustrates geometrically what RPA achieves on this synthetic dataset. The left panel (coloured by subject) shows each subject's covariance distribution before and after the three RPA steps (TLCenter $\rightarrow$ TLScale $\rightarrow$ TLRotate): subject-specific clusters that are initially dispersed across the 3-D projection collapse into a common region after alignment. The right panel (coloured by class) confirms that the two mental-state classes, which overlap before RPA, become linearly separable after alignment.

Quantitatively, TL alignment improves MDM from a mean AUC of 0.49 to 0.53 and—crucially—*substantially* benefits NCH, lifting it from 0.41 (below chance) to 0.49. Without alignment, NCH underperforms chance due to uncompensated domain shift. This confirms that RPA alignment is a necessary precondition for QAOA-based NCH to function in cross-subject settings.

B. QAOA Circuit Depth

We sweep the number of QAOA layers $p \in \{2, 3, 4, 5\}$ on a single-subject synthetic dataset (4 channels, 50 samples, 60 trials). Figure 8 shows that:

1) Both NCH variants (NaiveQAOA and Ulvi) maintain AUC ≈ 1.0 across all depths and incur negligible training time (< 1 s), since the NCH convex hull construction is the bottleneck.
2) QIOCE performance *degrades* from AUC ≈ 1.0 at $p = 2$ to ≈ 0.73 at $p = 5$, while training time grows from 30 s to 85 s. This suggests overfitting or optimization landscape hardening with deeper QIOCE circuits on small datasets.

Consequently, $p = 2$ layers is used in all MOABB experiments.

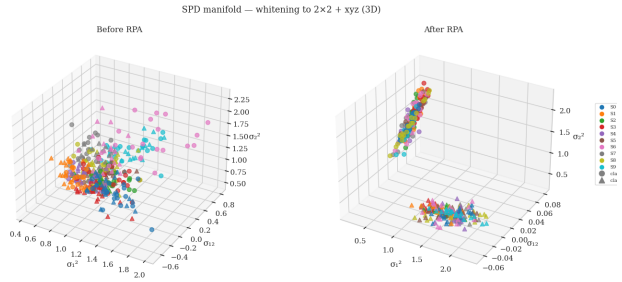

Fig. 7. Three-dimensional PCA projection of SPD covariance matrices on synthetic cross-subject EEG before and after Riemannian Procrustes Analysis (RPA). *Left*: points coloured by subject—RPA collapses each subject's cluster into a shared region, removing between-subject variability. *Right*: points coloured by class—after alignment, the two mental-state classes become linearly separable in the projected space.

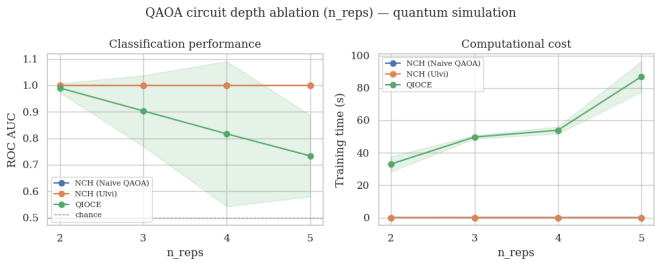

Fig. 8. QAOA circuit depth (n_reps) ablation. Left: ROC AUC. Right: training time. NCH variants (blue/orange) are stable across depths; QIOCE (green) degrades.

C. QIOCE Optimizer Comparison

We compare six optimizers on QIOCE using a 60-sample, 4-feature toy classification dataset (`make_classification`): L-BFGS-B, SLSQP, COBYLA, SPSA, NFT, and Anderson. Figure 9 summarizes ROC AUC, training time, and number of loss evaluations.

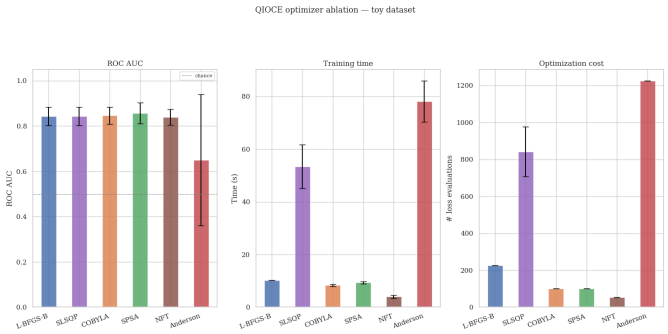

Fig. 9. QIOCE optimizer comparison. L-BFGS-B, SLSQP, COBYLA, SPSA, and NFT achieve similar AUC (≈ 0.84); Anderson achieves only 0.65 while consuming the most time (78 s) and evaluations (1,200).

Key findings:

- L-BFGS-B, SLSQP, COBYLA, SPSA, and NFT yield equivalent AUC ($\approx$0.84); NFT is the fastest (5 s, $\sim$60 evaluations).
- Anderson (the Ulvi optimizer) achieves the *lowest* AUC (0.65) with the *highest* computational cost (78 s, 1,200 evaluations) on this task. Its convergence curve (Fig. 10) shows slow descent with persistent high variance.
- NFT is chosen for the MOABB experiments as the best AUC/time trade-off optimizer.

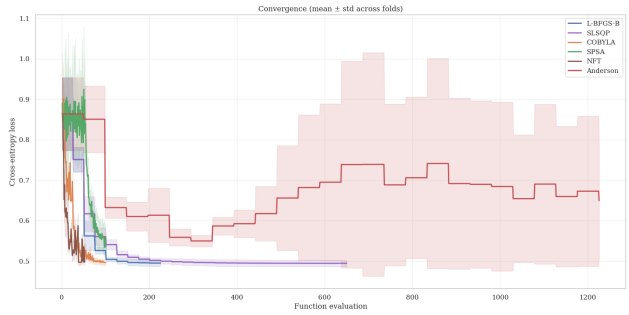

Fig. 10. Convergence curves (mean $\pm$ std across folds) for each optimizer. Gradient-based methods (L-BFGS-B, SLSQP) converge within 50 evaluations. Anderson converges slowly and exhibits high variance, never reaching the loss achieved by other methods.

a) Note on the Ulvi optimizer.: The Anderson/Ulvi optimizer was designed for QAOA-CV circuits with circular entanglement (*not* for the full QIOCE pipeline), so its poor performance on QIOCE is expected. In the NCH+QAOACV(Ulvi) setting, the optimizer operates on a different circuit topology and objective, and the Ulvi NCH variant outperforms Naive-QAOA by $+0.026$ mean AUC.

VI. Discussion

A. QuantumStateDiscriminator

The QuantumStateDiscriminator achieves the highest mean AUC (0.628) in the benchmark while being by far the fastest classifier (0.27 s mean training time), highlighting an important practical advantage. The absence of a statistically significant gap from MDM (0.613) suggests the quantum-theoretic PGM framework with Born-rule scoring is a viable alternative to Riemannian nearest-mean classification.

The key insight is that trace-normalized EEG covariance matrices are valid density matrices, so the quantum measurement formalism applies directly. The PGM couples all classes through ρ_{tot}, which is closer in spirit to LDA than MDM. Extending this approach with shrinkage regularization (Ledoit–Wolf normalization before trace normalization) or incorporating within-class structure into the POVM construction are natural directions.

B. QAOA-CV Ulvi Optimizer and NCH

The Ulvi circular-entanglement NCH variant consistently outperforms NaiveQAOA by approximately 0.026 mean AUC points, with no significant increase in training time. However, both variants remain significantly below classical baselines

(TS+LDA > NaiveQAOA, $p = 0.04$; MDM > Ulvi+TL, $p = 0.04$).

The ablation study (Section V-A) shows that RPA alignment is *necessary* for NCH to be reliable in cross-subject settings: without alignment, NCH falls below chance due to domain shift. Despite RPA, NCH with TL still underperforms MDM. This suggests the QAOA-CV optimization landscape is harder to navigate in high-dimensional EEG covariance space than in synthetic low-dimensional settings.

The QAOA circuit depth ablation confirms $p = 2$ layers is sufficient for NCH (performance saturates at 1.0 AUC on toy data for both variants).

C. IGL

IGL underperforms all classical baselines (mean AUC 0.522), with the gap to TS+LDA being statistically significant ($p = 0.04$, $d = 1.32$). This is surprising given that IGL operates on raw time series and theoretically adapts its intrinsic dimensionality to the data. Possible explanations include: (i) the 500 training epochs may be insufficient for the Variable Projection optimizer to converge on the relatively small EEG datasets; (ii) the Gaussian Green operator may not be well-suited to the temporal structure of resting-state EEG; (iii) cross-subject variability may exceed the capacity of a single shared kernel. Tuning `n_components`, `n_anchors`, and the kernel operator type per dataset is a potential avenue for improvement.

D. Transfer Learning

Counter-intuitively, RPA alignment degrades performance for most quantum pipelines on these datasets. For QIOCE, the whitening and tangent-space projection pipeline may already provide implicit alignment, making RPA redundant or harmful. For NCH, RPA improves synthetic cross-subject performance but does not recover the full performance gap to classical methods on real data. The significant advantage of MDM+TL over TS+LDA+TL suggests that RPA benefits Riemannian distance-based classifiers more than tangent-space methods.

E. Limitations

All quantum circuits are *simulated* classically; hardware noise and decoherence are not modeled. The benchmark uses a limited number of datasets and assumes the LWF covariance estimator is universally appropriate. The random seed at each run introduces variability in QAOA initialization; full reproducibility would require fixed seed across runs. [**TODO:** Add note on hardware-feasibility of circuit sizes.]

VII. Conclusion

We have presented a systematic cross-subject benchmark of 13 classical and quantum EEG classifiers on three MOABB resting-state datasets. Three new contributions were introduced and evaluated for the first time on real EEG: (i) QuantumStateDiscriminator, which achieves the highest mean AUC (0.628) at negligible computational cost by applying the Pretty Good Measurement directly on EEG density matrices; (ii) NCH+QAOA-CV with the Ulvi circular-entanglement optimizer, which outperforms the NaiveQAOA baseline while requiring significantly more work to match classical methods; (iii) IGL, which currently underperforms classical baselines on resting-state data.

The best overall configuration is QuantumStateDiscriminator (AUC 0.628, time 0.27 s), followed closely by QIOCE (0.626, ≈15 min) and MDM (0.613, 16 s). Classical baselines remain competitive, and no quantum method is yet statistically superior.

Future work includes: execution on near-term quantum hardware to account for noise; evaluation on motor imagery and SSVEP paradigms; and hybrid quantum-classical architectures that combine PGM-style feature scoring with Riemannian alignment.

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